Transjakarta:
Jakarta Public Bus Interconnectivity and Coverage

- ***Introduction***

As the world's **longest Bus Rapid Transit (BRT) system** , operates in a region projected to become the **largest metropolitan area by 2030**, ensuring **efficient interconnectivity and accessibility** is an **increasing challenge**. **Data-driven insights** are crucial to guide **future improvements**.

Stakeholder: **Government** and **State-owned Company Executives**

- ***Motivation***

As a regular Transjakarta user, I've seen its impact firsthand and contribute to it. This project examines how well it meets public needs and where it can improve, offering insights to support better transport planning in Jakarta and other growing cities.

- ***Aims***

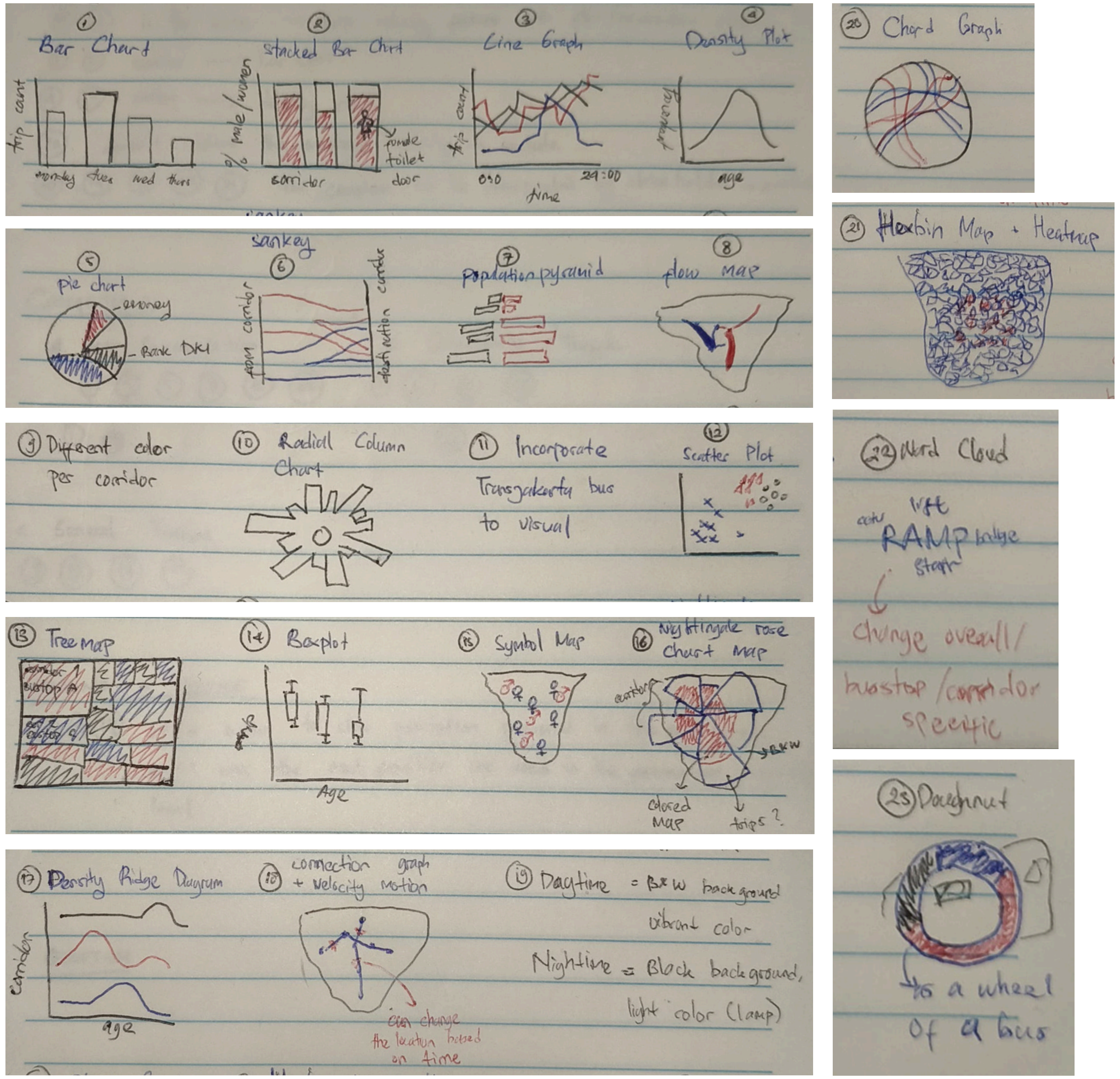
Explore and analyze the data to answer:

- How do bus stop usage patterns and traffic volumes affect route efficiency?
- What are the user profiles across routes, and how can this inform service improvements?
- Which bus stops require upgrades to accommodate increasing passenger demand?

Muhammad Iqbal
35092882

Title: *Transjakarta: Jakarta Public Bus Interconnectivity and Coverage*
Author: *Muhammad Iqbal*
Date: *22 May 2025*
Sheet: *1*
Task: *Brainstorming*

1. Ideas



2. Filter

6,10,14,15,13,16,20,21 are too complicated for stakeholders → *Exclude*
1 and **2** is similar → *Bar Chart*
8 and **10** is similar → *Flow/Connection Graph on top of Map*
4 and **17** is similar → *Density Chart*
5 and **23** is similar → *Pie Chart*

3. Categorize

A. User Profile: {2, 4, 5, 7, 12} → Visualize the proportion, behaviour, and characteristics of the users

B. Connectivity/Traffic: {2, 3, 8} → Shows bus traffic over time and location

C. Features: {9, 11, 18, 19} → Improve and refine the presentation and clarity of the visualizations

4. Combine and Refine

{2, 5} → Combine to compare 3+ categories and show aggregate comparisons.

{8, 9,18,19} →Flow maps with colored corridors (also as a filter), time filters (day/night) color change the background (gray and black), and stop-level connections with relative size indicate the tap-in and tap-out.

{2, 3,9,11} → Stacked bar and line charts to compare weekday vs weekend patterns over time, with different colors of corridor as divider and filter.

5. Question

- Does the visualizations sufficient to provide the insights to the questions?
- Is it effective enough to deliver with current combinations of visualizations?
- Does the filter is enough to give the stakeholders chance to analyze different insights?

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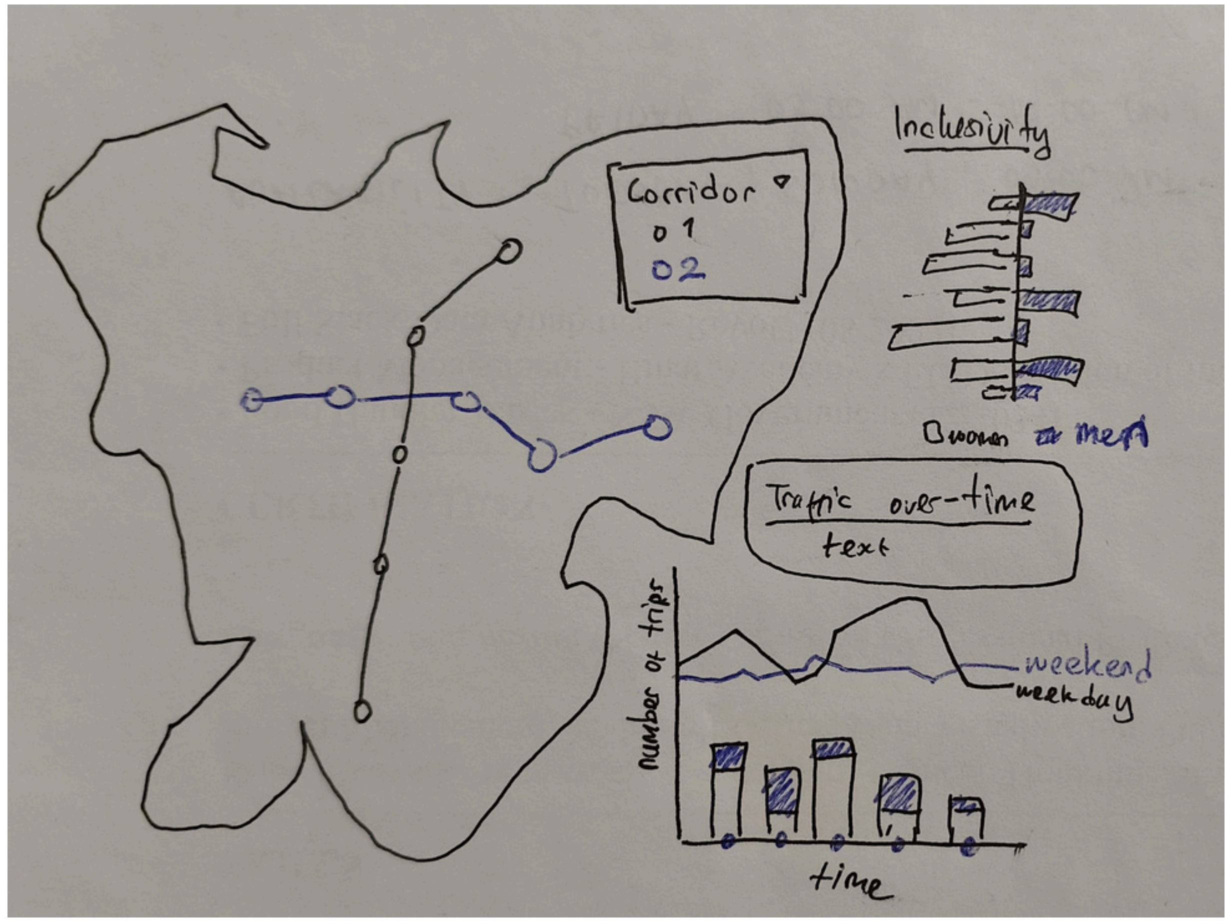
Task: *Alternative Designs - 1*

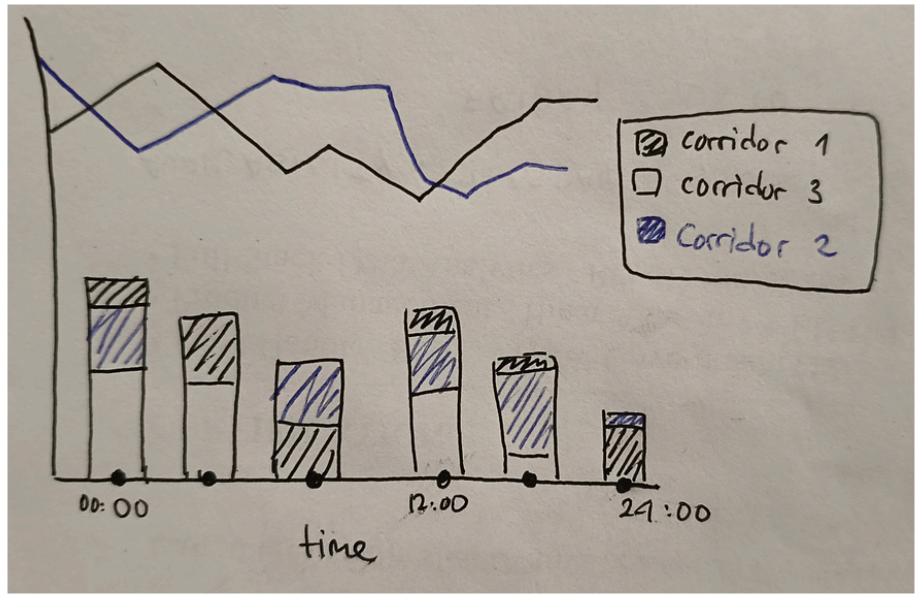
2. Layout



3. Operations

- Interactive map: Hovering over bus stops shows details (year built, accessibility, corridor, etc.).
- Legend click: Selecting a corridor in the legend filters all visualizations to highlight only that corridor (others dimmed). Click again to reset.
- Time focus: Clicking the x-axis (time) highlights specific time windows (e.g., evening rush). After 6 PM, a dark mode activates across all visuals to simulate night conditions.
- Reset: Clicking again on time/corridor restores default view.

4. Focus



- Highlights traffic trends across corridors and time, with emphasis on comparing weekday vs weekend usage.
- The combination of line and stacked bar charts gives both detailed and aggregated insights on corridor performance.

5. Discussion

(+) → Supports multi-perspective insights (time, location, corridor).
→ Geographic context makes corridor relevance easier to grasp.
→ Interactive features allow users to explore and extract tailored insights.

(-) → Multiple filters (corridor + time) could increase complexity for some stakeholders.
→ May require additional onboarding or tooltip hints to guide users in using filters effectively.

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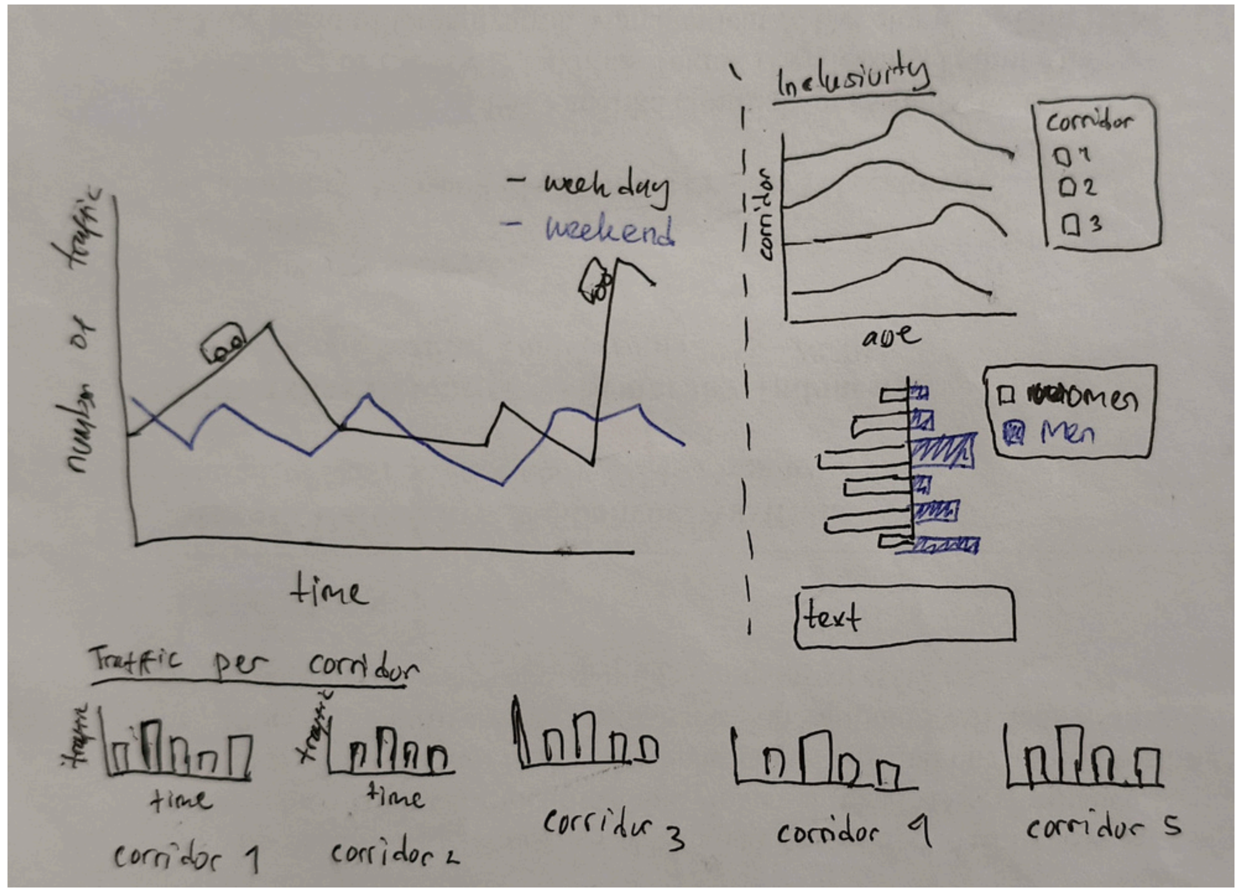
Sheet: *3*
Task: *Alternative Designs - 2*

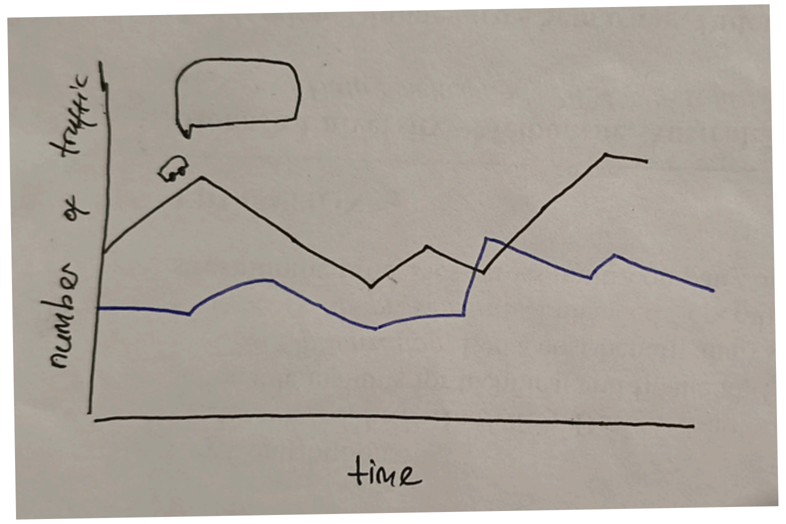
2. Layout



3. Operations

- Interactive time filtering: Users can click the bus icon on the line chart to filter data by a selected time range. Other charts will automatically update.
- Legend-based filtering: Each chart's legend can be clicked to isolate specific corridors or days. Inactive categories fade to gray (lower opacity).
- Hover interactions: Tooltips display exact values and percentages for each bar/line segment.

4. Focus



- Designed to communicate real-time trends and usage patterns to stakeholders.
- Emphasizes the comparison between weekday and weekend traffic across corridors.
- Supports planning decisions by visually breaking down trends over time and per corridor.

5. Discussion

(+) → Rich set of visualizations to address multiple questions and insights
→ Separated bar charts by corridor enhance clarity and comparability.
→ Clear trend differentiation suitable for timely report.

(-) → Lack of geographic map might make it difficult for some users to understand corridor locations.
→ Time filter via bus icon could be unclear without visual cues or tooltips.
→ May require onboarding or user guidance to fully utilize all filtering functions.

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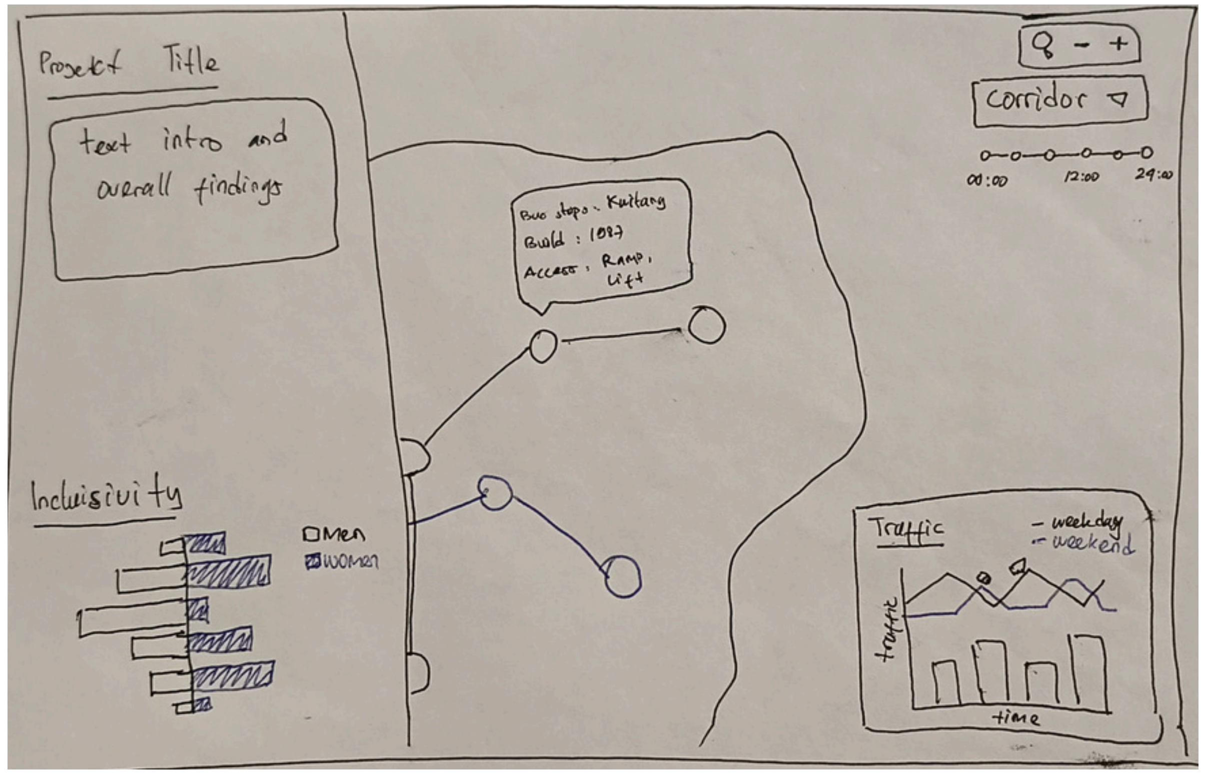
Sheet: *4*
Task: *Alternative Designs - 3*

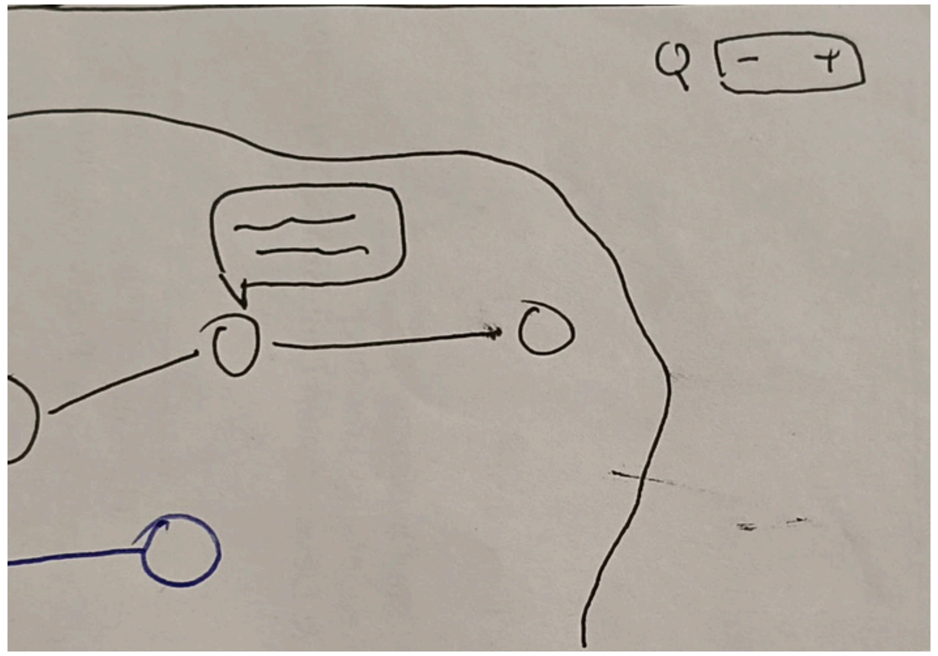
2. Layout



3. Operations

- User can zoom-in and zoom-out to make to choose between focused on each of the bus stops closer or overall glance.
- Hovering across dot in connection chart will show tooltips that provides more details of each bus stops (name, corridor, built year, accessibility)
- There are corridor (drop-down) and time range filter to filter out the data to be more focused based on the user needed, and the other categories in other chart will be grayed-out and lower opacity.

4. Focus



- This design emphasizes bus stop-level insights using a map-centered visualization.
- By combining location context with interactive filters (e.g., time range, corridor selection), the design supports both specific investigation and holistic overview, helping stakeholders identify stop-specific issues and broader patterns.

5. Discussion

(+) → The centered map helps users understand where each bus stop operates, which improves spatial reasoning.

→ Zooming and filtering enable flexible exploration based on user needs, making the design more adaptable for various stakeholders.

(-) → When charts are too small or too dense, it may become difficult for non-technical users to extract meaningful insights without guidance.

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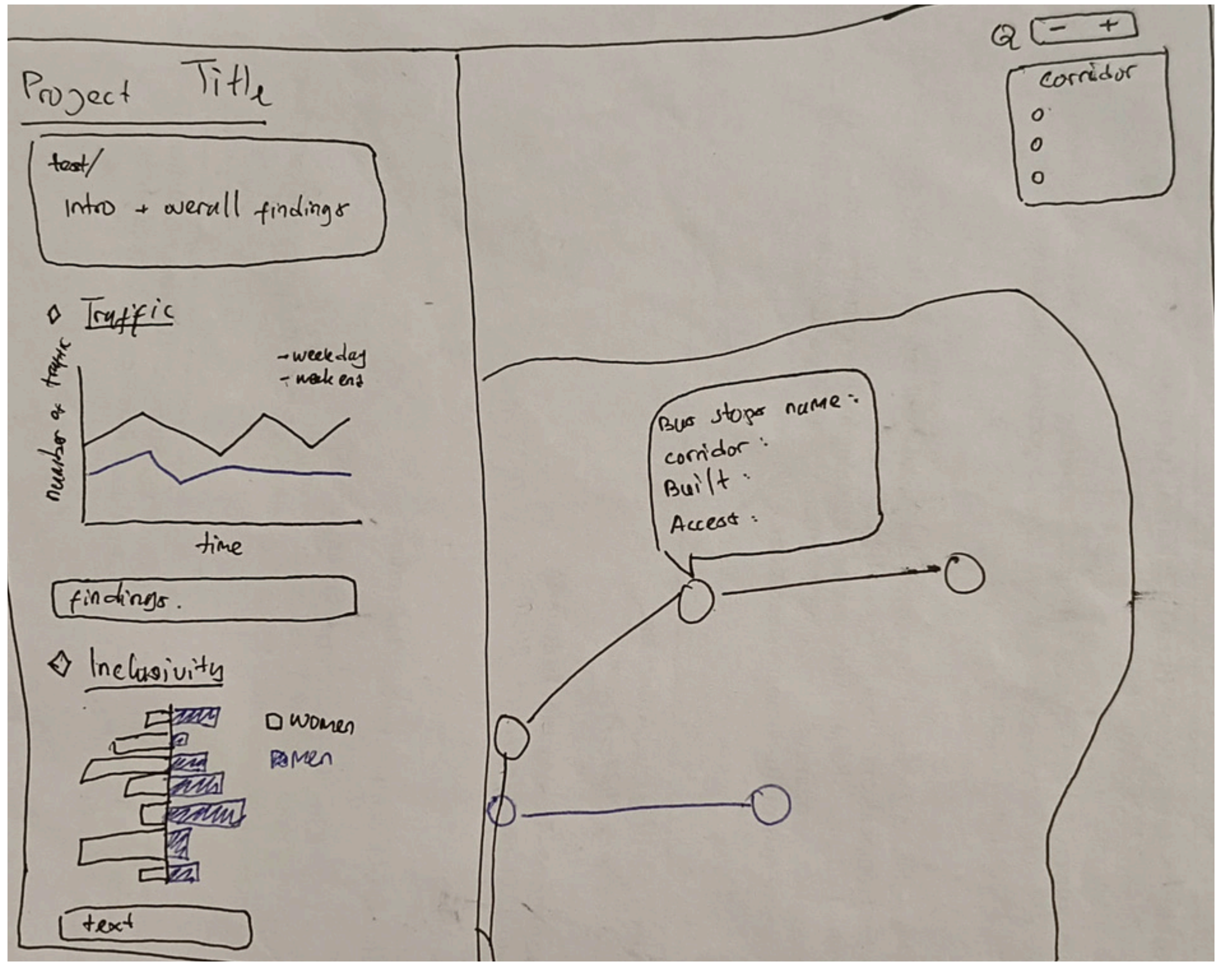
Sheet: *5*
Task: *Realisation*

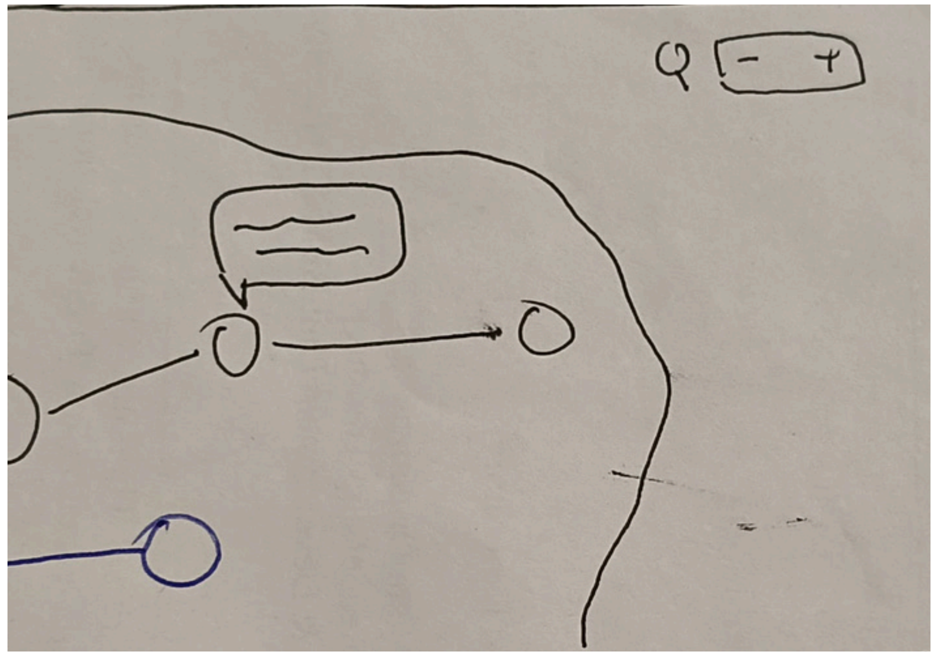
2. Layout



3. Operations

- Users can zoom in/out to view individual stops or the overall connection bus stops chart.
- Hover tooltips: Display detailed stop information (name, corridor, year built, accessibility).
- Users can filter data by corridor through the legend or by time range, with all charts and the map updating dynamically. Non-selected elements are dimmed, ensuring a consistent and focused view across the dashboard.

4. Focus



- Enables users to explore bus stop-level insights in both local detail and citywide context.
- Supports planning and policy decisions by revealing stop-specific performance and broader patterns across corridors.

5. Details

Datasets → A combined dataset consisting 2 datasets: generated datasets from kaggle with references of real historical traffic data of Transjakarta April 2023, and Bus Stops details data from Greater Jakarta Government OpenData portal of 2023. Data cleaned and wrangled using R.

Dependencies → D3 in JavaScript

Estimations → Cost : -

→ Time: 29 May 2025 (Overall Initial Layout and Static Chart)

4 June 2025 (Filter and Interactions)

9 June 2025 (Finishing)